

Recursive Rendering of 2D Images for Accurate Pose Estimation in a 3D Mesh Map

Yohei Hanaoka
KDDI Research, Inc.
Saitama, Japan
yo-hanaoka@kddi.com

Suwichaya Suwanwimolkul
KDDI Research, Inc.
Saitama, Japan
xsu-suwanui@kddi.com

Satoshi Komorita
KDDI Research, Inc.
Saitama, Japan
sa-komorita@kddi.com

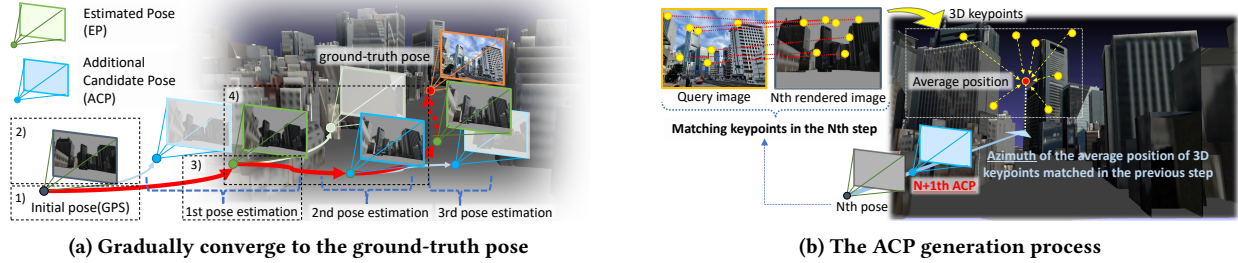


Figure 1: Our proposed recursive rendering and camera pose estimation in a 3D city mesh map

ABSTRACT

Image-based pose estimation generally needs 3D map generation, which is costly in practice. In this paper, we propose a method to accurately estimate the pose by recursively rendering 2D images from a generic 3D city mesh map and updating the pose.

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1 INTRODUCTION

Due to its high precision and low facility requirements, image-based visual localization is an essential part of various 3D vision technologies including navigation, autonomous driving, and augmented reality. Image-based visual localization starts from collecting images to construct the 3D map in a database. The 3D map consists of 3D point clouds associated with 2D local features, *i.e.*, the keypoints and descriptors of the database images. The 2D local features are the essential part in establishing the 2D correspondences between the query image of the camera to be localized and the database images in the 3D map. Given the 3D point clouds and the 2D keypoints in query images, the camera pose, *i.e.*, the position and rotation, can be estimated. Minor decrease in visual similarity such as the illumination environment or viewpoints changes can lower the correspondences, reducing the pose accuracy, so highly robust 2D local features, *e.g.*, [Revaud et al. 2019], have been recently improved.

However, 3D point clouds dedicated to 2D local features must be generated in prior in visual localization. In addition, we need to re-create the map once we change the local features for the higher performance or to meet a certain criteria for each environment, *e.g.*, smartphone and vehicles, which requires different features to maximize the performance under each computer resource. In practice, the 3D map creation is costly and laborious.

To address this problem, we propose a solution for estimating the pose on a generic 3D map that does not rely on specific 2D local features. We use a 3D mesh city map as the generic 3D map, which is a publicly available map for urban development. We estimate the pose of a query image by establishing the correspondences between the query image and the images rendered from the 3D mesh. However, because of its generality, the resolution of texture is often low, especially in open 3D mesh city maps. The significant difference in visual similarity between a query image and images rendered from a 3D mesh map results in low matching accuracy. Thus, we employ the advantage in the geometry of the rendered images, where we can render the new image whose the camera pose is moved closer to the query image to improve the correspondences. Our proposed method recursively renders images and updates the pose to achieve accurate pose estimation.

2 PROPOSED METHOD

2.1 Recursive rendering

Our goal is to estimate the accurate pose corresponding to the query image in the 3D mesh map by comparing the query image and an image rendered from the 3D mesh map. In general, the closer the pose of the rendered image to the ground-truth (GT) pose of the query image the greater the accuracy of the estimated pose because the viewpoint changes between the two images are small. Furthermore, we hypothesize that the estimated pose will be better than the pose of the rendered image used for estimation. Based on the hypothesis, we propose a method for updating the estimated pose with the goal of making it closer to the GT after initializing the

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Table 1: Experimental results

Methods	1m,2°(%)	2.5m,5°(%)	5m,10°(%)	MLE (m)	MAE (°)
geo-tags	0.0	1.6	14.3	5.56	16.17
Pre-Rendering	17.5	38.1	55.6	3.78	1.56
One-Time-Rendering	28.6	44.4	52.4	3.52	1.07
<i>Ours w/o ACP</i>	36.5	47.6	50.8	3.52	1.79
<i>Ours w/ ACP</i>	41.3	60.3	66.7	1.31	0.51

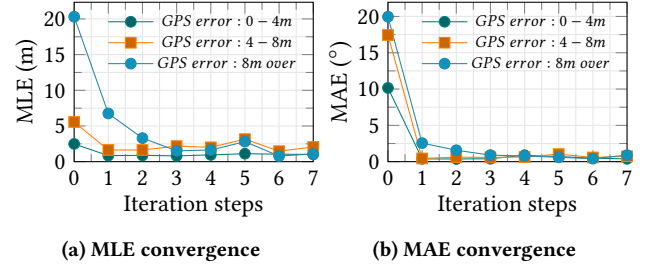
pose. This process is shown in Figure 1a: 1) obtain an initial pose using GPS and magnetometer sensors, denoted as "geo-tags" here; 2) render an image from a 3D mesh map at the initial pose; 3) compare the query image and the rendered image, and estimate the pose of the query image solving a perspective-n-point (PnP) problem; 4) render an image at the estimated pose and estimate the next pose of the query image again. We repeat step 4 recursively until the difference between each estimated position becomes sufficiently small. In Figure 1, the 3D city mesh map PLATEAU [Ministry of Land, Infrastructure, Transport and Tourism 2020] was used.

2.2 Improvement with additional candidate pose

In some cases, we obtain unsuitable rendered images due to the occlusions during the recursive rendering process. This incurs position divergence and erroneous results. To avoid producing an erroneous image and to improve robustness, we generate an additional candidate pose (ACP) to be matched differently from the estimated pose (EP) at each step. Then, we select the better one by matching the results between the query image and images rendered at both poses. Here, we generate an ACP by shifting from the previously estimated pose to the azimuth, which is oriented to the average position of 3D keypoints matched in the previous step because the camera can continue to see the same 3D points. The moving distance is about 10% of the distance between the previously estimated location and the average position of 3D keypoints. Figure 1b shows the process of ACP generation.

3 EXPERIMENT

We evaluated our proposal using a 3D mesh city map [Ministry of Land, Infrastructure, Transport and Tourism 2020] of about 2 km² and 63 query images taken in the same area of the map. We published the dataset in [Hanaoka et al. 2022]. We used the LoFTR [Sun et al. 2021] framework for image matching, due to the fact that it performed well in our preliminary experiments. This method can establish the correspondence despite images having low similarity by performing dense image matching. The recursive process terminates when it converges or reaches the 10th iteration without convergence. Based on our preliminary experiment, we set the threshold of the difference between subsequent estimated positions to 1m for determining the convergence. Note that we rendered three initial images by rotating horizontally around the geo-tags by ± 30 degrees and used the one with the best matching result as the initial image due to the significant errors of magnetometer sensors in urban environments [Hölzl et al. 2013]. We evaluated our proposed method on two patterns: with and without the ACP, denoted as *w/ ACP* and *w/o ACP*. We also performed three reference experiments. The first is "geo-tags," which is just the initial position by geo-tags. The second is "Pre-Rendering," which prepares images

**Figure 2: Error convergence of the estimated pose with ACP**

in advance and uses the rendered image whose pose is closest to the geo-tags of the query image. We prepared 5,362 images as the database images at intervals of 10m in the 3D mesh area at a location with good visibility as the main street. The rendering intervals refer to the standard used in [Torii et al. 2015]. Pre-Rendering is an approach equivalent to existing image-based visual localization. The third is "One-Time-Rendering," which uses only the initial rendered image as a simple baseline method in a renderable environment.

Table 1 shows the experimental results. The evaluation metrics are localization errors (LE)(m) and angular errors (AE)(°) between the estimated pose and the GT. We report the accuracy rate (%) of the errors within three thresholds (1m, 2°), (2.5m, 5°), (5m, 10°) and the median of LE (MLE) and AE (MAE). The results show that *Ours w/ ACP* performs best in terms of the accuracy rate. One-Time-Rendering results have almost the same MLE as Pre-Rendering. In contrast, *Ours w/ ACP* reduced the MLE by 65.3% compared to Pre-Rendering. Figure 2 shows the convergence of MLE and MAE for the initial error size due to GPS. Here, 0th step means geo-tags error. We excluded trials for which sufficient corresponding points for pose calculation were not obtained. Although the initial error is large, the error decreases with each iteration. Note that the error does not decrease monotonically after the third iteration due to the inclusion of some trials with divergent position values. These results indicate our hypothesis is somewhat supported.

4 CONCLUSION

We present a method to accurately estimate the pose by recursively rendering images from a 3D mesh city map and updating the pose. The results of the experiments demonstrated that our proposal was the best in terms of accuracy.

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